1.3.1. Prove Lem 1.3.1 : $\begin{cases} \Lambda = w_1\mathbb{Z} \oplus w_2\mathbb{Z} \\ \Lambda' = w_1'\mathbb{Z} \oplus w_2\mathbb{Z}. \end{cases}$

$w_1/w_2,\ w_1'/w_2' \in \mathcal{H}$.

Then $\Lambda' = \Lambda$ iff $\begin{pmatrix} w_1' \\ w_2' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$, $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$.

Pf: $\Leftarrow$: $\Lambda' \subset \Lambda \Leftarrow \begin{cases} w_1' = a w_1 + b w_2 \\ w_2' = c w_1 + d w_2 \end{cases}$.

$\Lambda' \supset \Lambda \Leftarrow \begin{cases} w_1 = a' w_1' + b' w_2' \\ w_2 = c' w_1' + d' w_2' \end{cases}$ $\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}$.

$\Rightarrow$: $\begin{cases} w_i = x_i + y_i i \\ w_i' = x_i' + y_i' i \end{cases}$ $\Rightarrow$ this is exactly why image is still a lattice.

$\Lambda' = \Lambda \Rightarrow \begin{pmatrix} w_1' \\ w_2' \end{pmatrix} = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\in GL_2(\mathbb{Z})}\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1' & y_1' \\ x_2' & y_2' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}$.

$w_1/w_2 \in \mathcal{H} \Rightarrow \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} < 0$ $\quad w_1'/w_2' \in \mathcal{H} \Rightarrow \begin{vmatrix} x_1' & y_1' \\ x_2' & y_2' \end{vmatrix} < 0$.

Thus ~~det~~ $\begin{vmatrix} a & b \\ c & d \end{vmatrix} > 0$. as is in $GL_2(\mathbb{Z})$, is in ~~GL_2~~ $SL_2(\mathbb{Z})$.

1.3.2. Prove Cor 1.3.3. $\varphi : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ holomorphic.

$\varphi(z+\Lambda) = mz + b + \Lambda'$, $m\Lambda \subset \Lambda'$.

Then (1) φ group homomorphism $\Leftrightarrow$ (2) $b \in \Lambda'$. thus $\varphi(z+\Lambda) = mz + \Lambda'$

$\Leftrightarrow$ (3) $\varphi(0) = 0$.

Pf: (3) $\Leftrightarrow$ (2) $\varphi(0) = \varphi(\Lambda) = b + \Lambda' = \Lambda' \Rightarrow b \in \Lambda'$.

$b \in \Lambda' \Rightarrow \varphi(0) = \varphi(\Lambda) = \Lambda'$ true.

(1) $\Rightarrow$ (3) $\varphi(0) = \varphi(z+\Lambda) - \varphi(z+\Lambda) = \Lambda' - \Lambda' = \Lambda'$ true.

(2) $\Rightarrow$ (1) $\varphi(z_1 + z_2 + \Lambda) = m(z_1+z_2) + \Lambda' = mz_1 + \Lambda' + mz_2 + \Lambda'$

$= \varphi(z_1 + \Lambda) + \varphi(z_2 + \Lambda)$.

1.3.3 (a) Show that Weil pairing is independent of the choice of basis $\{w_1, w_2\}$, $w_1/w_2 \in \mathcal{H}$.

Pf: If $\begin{pmatrix} P \\ Q \end{pmatrix} = \gamma_1 \begin{pmatrix} w_1/N+\Lambda \\ w_2/N+\Lambda \end{pmatrix} = \gamma_2 \begin{pmatrix} w_1'/N+\Lambda \\ w_2'/N+\Lambda \end{pmatrix}$.

then $\Lambda = w_1\mathbb{Z} \oplus w_2\mathbb{Z} = w_1'\mathbb{Z} \oplus w_2'\mathbb{Z}$

$\Rightarrow \begin{pmatrix} w_1' \\ w_2' \end{pmatrix} = \gamma \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$ $\gamma \in SL_2(\mathbb{Z})$.

Thus $\gamma_1 = \gamma_2 \gamma$. $\Rightarrow e_N(P,Q) = e^{\frac{2\pi i \det \gamma_1}{N}} = e^{\frac{2\pi i \det \gamma_2}{N}}$.

(b) Show that Weil pairing is bilinear, alternating and nondegenerate.

Pf: Bilinear: ~~$\begin{pmatrix} P_1+P_2 \\ Q \end{pmatrix} = \gamma_1 + \gamma_2$~~ If $\begin{pmatrix} P_i \\ Q \end{pmatrix} = \begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix} \begin{pmatrix} w_1/N+\Lambda \\ w_2/N+\Lambda \end{pmatrix}$ $i=1,2$

then $\begin{pmatrix} P_1+P_2 \\ Q \end{pmatrix} = \underbrace{\begin{pmatrix} a_1+a_2 & b_1+b_2 \\ c & d \end{pmatrix}}_{=\gamma} \begin{pmatrix} w_1/N+\Lambda \\ w_2/N+\Lambda \end{pmatrix}$.

$\det \gamma = \det \gamma_1 + \det \gamma_2$.

Thus $e_N(P_1+P_2, Q) = e_N(P_1, Q) \cdot e_N(P_2, Q)$. same for Q.

Alternating: $\begin{pmatrix} P \\ Q \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} w_1/N+\Lambda \\ w_2/N+\Lambda \end{pmatrix} \Rightarrow \begin{pmatrix} Q \\ P \end{pmatrix} = \underbrace{\begin{pmatrix} c & d \\ a & b \end{pmatrix}}_{=\gamma_2} \begin{pmatrix} w_1/N+\Lambda \\ w_2/N+\Lambda \end{pmatrix}$.

$\det \gamma_2 = -\det \gamma_1$.

Nondegeneracy: $e_N(Q, P) = e_N(P, Q)^{-1}$.

$\forall P \neq 0$, $\begin{pmatrix} P \\ Q \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} w_1/N+\Lambda \\ w_2/N+\Lambda \end{pmatrix}$ $a \cdot b \neq 0$.

If $a \neq 0$. Let $d \neq 0$. $\exists Q$ $e_N(P,Q) \neq 1$.

Equivalently, $Q \mapsto e_N(P,Q)$ nontrivial character of $E[N]$.

(1.3.3) (c) Show that Weil pairing is compatible with N.

i.e.

$$\begin{tikzcd} E[dN] \times E[dN] \arrow[r] \arrow[d, "d(\cdot,\cdot)"'] \arrow[dr, phantom, "\circlearrowright"] & \mu_{dN} \arrow[d, "\cdot^d"] \\ E[N] \times E[N] \arrow[r] & \mu_N \end{tikzcd}$$

Pf: left bottom: $(P, Q) \mapsto (dP, dQ) \mapsto e^{\frac{2\pi i \det \gamma}{N}}$

~~top~~ ~~since by bilinearity,~~

~~$e_N(dP, dQ) = e_N(P, Q)$~~

$$\begin{cases} P = \frac{a_1 w_1 + b_1 w_2}{dN} + \Lambda \\ Q = \frac{a_2 w_1 + b_2 w_2}{dN} + \Lambda \end{cases}$$

top right: $(P, Q) \mapsto e^{\frac{2\pi i \det \gamma}{dN}}$.

$$\begin{cases} dP = \frac{a_1 w_1 + b_1 w_2}{N} + \Lambda \\ dQ = \frac{a_2 w_2 + b_2 w_2}{N} + \Lambda \end{cases}$$

Thus $\cdot^d : \mu_{dN} \to \mu_N$. $\quad d(\cdot,\cdot) : E[dN]^2 \to E[N]^2$

$x \mapsto x^d$. $\quad (P, Q) \mapsto (dP, dQ)$

(d) Let $m\Lambda = \Lambda'$. Show $\mathbb{C}/\Lambda \xrightarrow{\sim} \mathbb{C}/\Lambda'$ preserves the Weil pairing. $z + \Lambda \mapsto mz + \Lambda'$

Pf:

$$P = \frac{a_1 w_1 + b_1 w_2}{N} + \Lambda,$$

$$Q = \frac{a_2 w_1 + b_2 w_2}{N} + \Lambda.$$

$$\varphi(P) = \frac{a_1(m w_1) + b_1(m w_2)}{N} + \Lambda', \quad \varphi(Q) = \frac{a_2(m w_1) + b_2(m w_2)}{N} + \Lambda'$$

$$e_N(P, Q) = e^{\frac{2\pi i (a_1 b_2 - b_1 a_2)}{N}}, \quad e'_N(\varphi(P), \varphi(Q)) = e^{\frac{2\pi i (a_1 b_2 - b_1 a_2)}{N}}$$

$$e_N = e'_N.$$

$$\begin{pmatrix} \varphi(P) \\ \varphi(Q) \end{pmatrix} = \gamma \begin{pmatrix} \frac{m w_1}{N} + \Lambda' \\ \frac{m w_2}{N} + \Lambda' \end{pmatrix}.$$